

Experiment 7: Random Sampling and Sampling Distribution

AIM:

To explore random sampling from a population, observe the properties of sampling distributions across varying sample sizes using Python, and demonstrate the Central Limit Theorem.

CODE:

```
import numpy as np
import matplotlib.pyplot as plt

# Step 1: Generate a population (Normal Distribution)
population_mean = 50
population_std = 10
population_size = 100000

# Set seed for reproducible output
np.random.seed(42)
population = np.random.normal(population_mean, population_std, population_size)

# Step 2: Perform Random Sampling
sample_sizes = [30, 50, 100] # Different sample sizes
num_samples = 1000           # Number of samples per size

sample_means = {}

for size in sample_sizes:
    sample_means[size] = []
    for _ in range(num_samples):
        sample = np.random.choice(population, size=size, replace=False)
        sample_means[size].append(np.mean(sample))

# Print summary statistics of sampling distributions
print(f"Population Mean: {np.mean(population):.4f}")
print(f"Population Standard Deviation: {np.std(population):.4f}\n")

for size in sample_sizes:
    s_mean = np.mean(sample_means[size])
    s_std = np.std(sample_means[size])
    expected_std = population_std / np.sqrt(size)
```

```

print(f"Sample Size {size}:")
print(f" Mean of Sample Means: {s_mean:.4f}")
print(f" Std Dev of Sample Means (Standard Error): {s_std:.4f} (Theoretical SE:
{expected_std:.4f})")

# Step 3: Plotting Sampling Distributions
plt.figure(figsize=(12, 8))

for i, size in enumerate(sample_sizes):
    plt.subplot(len(sample_sizes), 1, i+1)
    plt.hist(sample_means[size], bins=30, alpha=0.7, color='skyblue', edgecolor='black',
label=f'Sample Size {size}')
    plt.axvline(np.mean(population), color='red', linestyle='dashed', linewidth=1.5,
label='Population Mean')
    plt.title(f'Sampling Distribution of the Mean (Sample Size {size})')
    plt.xlabel('Sample Mean')
    plt.ylabel('Frequency')
    plt.legend()

plt.tight_layout()
plt.savefig('sampling_distribution_output.png', dpi=300)
plt.show()

```

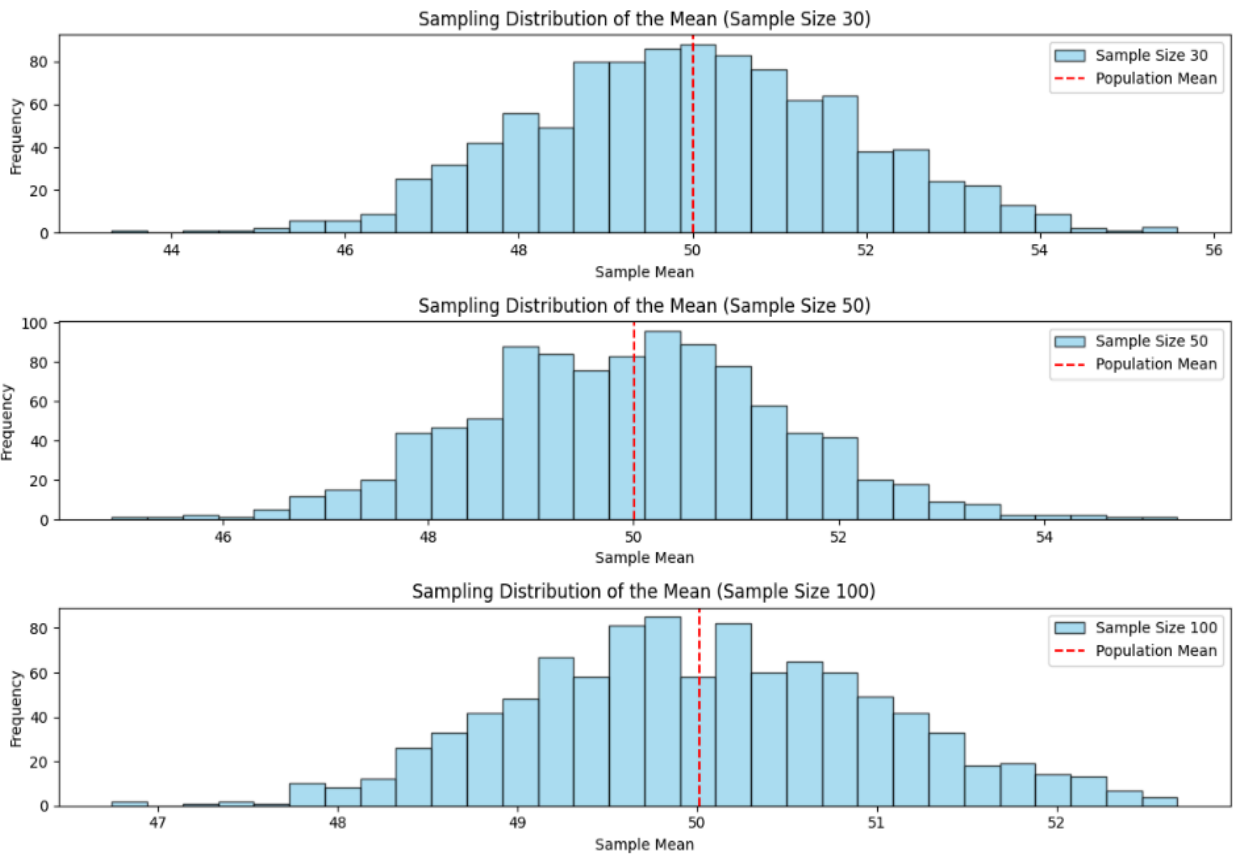
OUTPUT:

```

Population Mean: 50.0097
Population Standard Deviation: 10.0090

Sample Size 30:
  Mean of Sample Means: 49.9805
  Std Dev of Sample Means (Standard Error): 1.8564 (Theoretical SE: 1.8257)
Sample Size 50:
  Mean of Sample Means: 49.9567
  Std Dev of Sample Means (Standard Error): 1.4899 (Theoretical SE: 1.4142)
Sample Size 100:
  Mean of Sample Means: 50.0236
  Std Dev of Sample Means (Standard Error): 1.0001 (Theoretical SE: 1.0000)

```



RESULT:

The experiment on Random Sampling and Sampling Distribution was executed successfully using Python.